

The following insights and clarifications are offered in response to the material presented in “Probably some practical advice when using Flood Frequency Analysis utilities” by Lindsay Millard [1]. This paper makes a useful and informative contribution to the community of practice. I am glad to see people are using our software and taking a deeper dive into the details.

Replicating FLIKE Results with RMC-BestFit

The Millard paper suggests that many practitioners cannot replicate FLIKE [2] results using RMC-BestFit. The differences between FLIKE and RMC-BestFit results are mostly due to different prior distributions. The default prior in RMC-BestFit is a weakly informative uniform distribution for each parameter. Whereas FLIKE uses an objective, noninformative Jeffreys’ prior for scale parameters $f(\sigma) \propto \frac{1}{\sigma}$. In practice, the difference between the two approaches is negligible especially when considering the uncertainty in the estimate. The differences will rarely have any impact on our understanding of the flood hazard, the associated risks, or on our investment decisions. In most applied contexts, there is no clear advantage to using an objective prior like Jeffreys’ when sufficiently vague priors like a uniform distribution will suffice. Additional details are provided in the RMC-BestFit verification report (<https://www.rmc.usace.army.mil/Software/RMC-BestFit/>) [3].

The current version of RMC-BestFit does not directly support a Jeffreys’ prior. However, we can obtain results equivalent to FLIKE by approximating a Jeffreys’ prior in RMC-BestFit. We are considering adding Jeffreys’ prior for scale parameters as an option in a future version.

In the meantime, we can approximate a Jeffrey’s prior by changing the prior for the scale parameter to be a Gamma distribution. This technique was used to obtain the results portrayed in Figures 55 through 59 of the RMC-BestFit verification report (<https://www.rmc.usace.army.mil/Software/RMC-BestFit/>). This is not clearly stated in the verification report, and we apologize for any confusion.

For Example #3 in Book 3 of ARR 2019 [4], the Gamma prior was set to have a scale parameter equal to 2 and a shape parameter equal to 0.5. Figure 1 shows the required RMC-BestFit inputs.

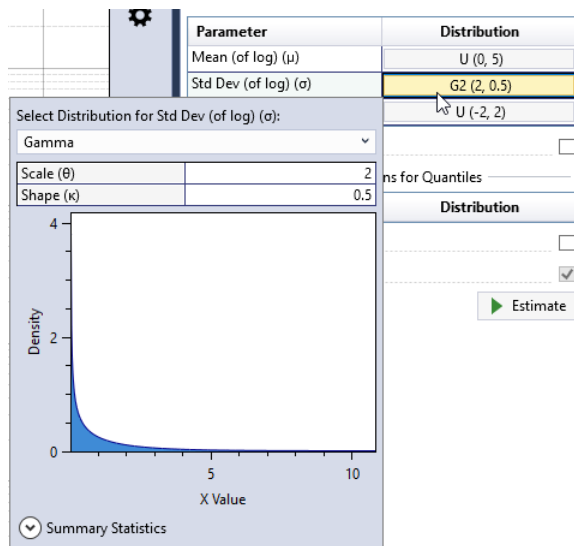


Figure 1 – Gamma prior meant to behave like Jeffreys’ prior for Example #3.

Results for Example #3 in Book 3 of ARR 2019 are summarized in the table and figures below. The difference in results is going to be negligible for most practical applications considering that the range of uncertainty spans several orders of magnitude. Generally speaking, for all Bayesian software, results will never be exactly identical due to Monte Carlo sampling error and potential differences in interpolation and rounding schemes. For all practical purposes, the results shown in Table 1 are the same. Figure 2 demonstrates the comparison between the FLIKE v5.306 results and RMC-BestFit with the Gamma Prior for Standard Deviation, while Figure 3 demonstrates the comparison between the FLIKE v5.306 results and the RMC-BestFit default Uniform Prior for Standard Deviation, as included in the Millard paper.

Table 1 - Comparison of results for Example #3.

Flow (cms)	Expected Exceedance Probability		
	FLIKE v5.306	RMC-BestFit (Gamma Prior for Standard Deviation)	RMC-BestFit (Default Uniform Prior for Standard Deviation)
3,928	0.101 (1 in 9.88)	0.103 (1 in 9.70)	0.108 (1 in 9.24)
12,786	0.0233 (1 in 43.0)	0.0241 (1 in 41.4)	0.0268 (1 in 37.3)
29,021	0.00843 (1 in 118.7)	0.00873 (1 in 114.6)	0.00999 (1 in 100.1)
47,033	0.00481 (1 in 207.9)	0.00499 (1 in 200.4)	0.00579 (1 in 172.7)

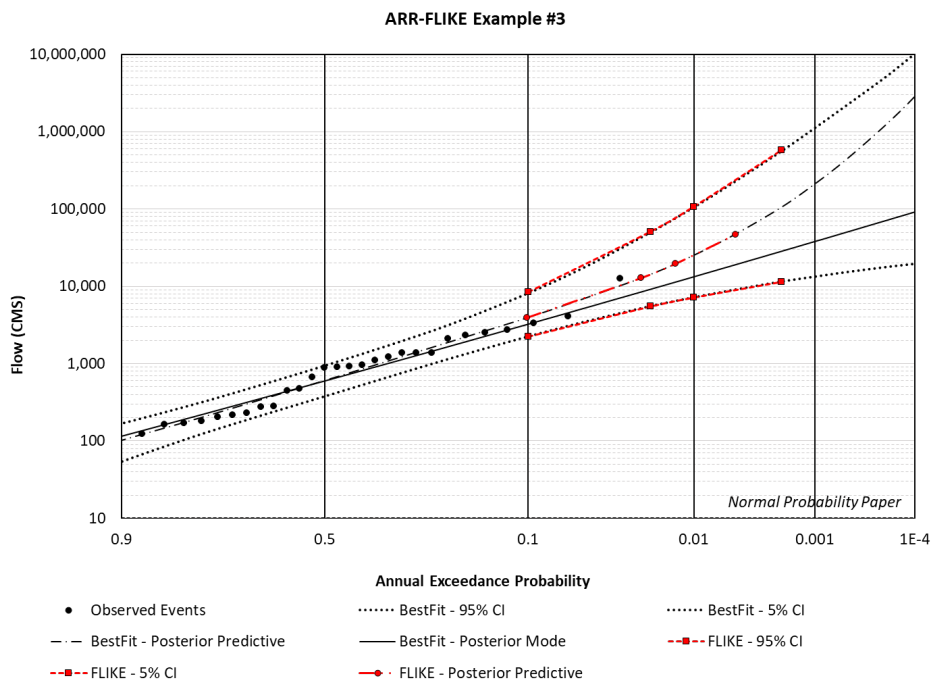


Figure 2 - Comparison Between FLIKE and RMC-BestFit (Gamma Prior for Standard Deviation) for Example #3.

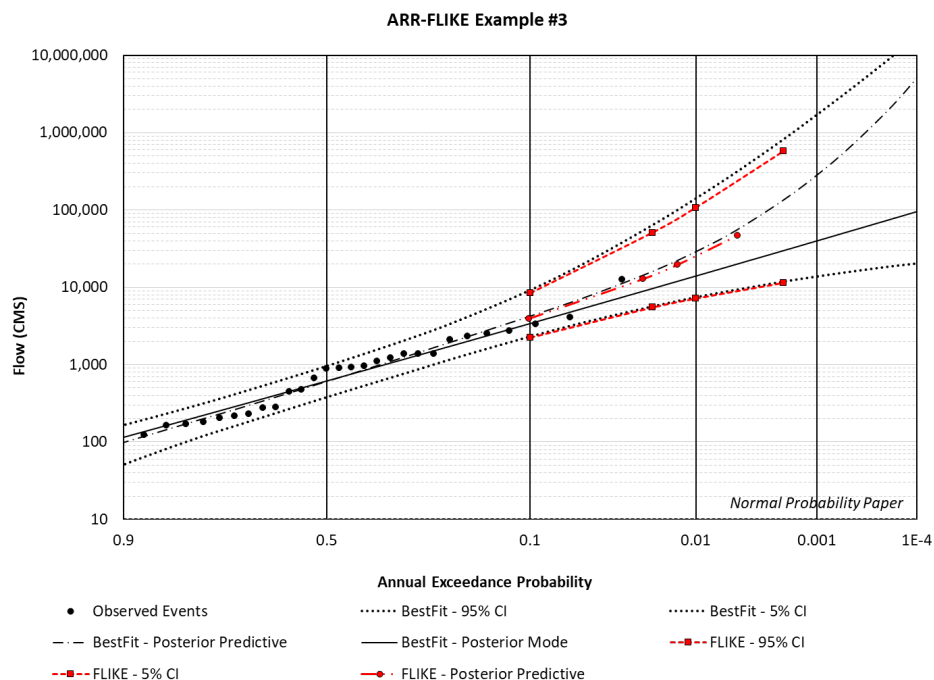


Figure 3 - Comparison Between FLIKE and RMC-BestFit (Uniform Prior for Standard Deviation) for Example #3.

Right (Binomial) Censored Data in RMC-BestFit

The Millard paper suggests that RMC-BestFit does not support right (binomial) censored data. RMC-BestFit accommodates right censored data using flow intervals. We can obtain results equivalent to FLIKE using flow intervals with the number of intervals equal to the number of exceedances; the lower value equal to the censoring threshold, and the upper value equal to a number that is equivalent to +infinity (e.g., 10,000,000).

For Example #4 in Book 3 of ARR 2019, the one flood exceedance above the 1955 flood that occurred between 1820 and 1937 can be entered as right censored data in RMC-BestFit using a flow interval with the year equal to 1820, the lower value equal to 12,526, the most likely value equal to 12,527, and the upper value equal to 10,000,000. The specific year entered for the flow interval does not matter in this case. It only matters that we input the correct number of exceedances using a separate flow interval for each exceedance.

For consistency, we recommend entering the flow interval(s) starting at the first year of the applicable time period. The software also requires that the most likely value be greater than the lower value. We typically enter a most likely value that is nominally greater than the lower value unless we have specific information about the flood magnitude. The upper value must be a sufficiently large numerical value that is equivalent to +infinity. A value of 10 million should suffice in most cases. The remaining years from 1821 to 1937 are entered as a perception threshold with a value of 12,526. The following screenshots show the required RMC-BestFit inputs. The figure that follows shows a comparison between the FLIKE and RMC-BestFit results. For all practical purposes, they are the same. Figure 4 demonstrates these input values in RMC-BestFit.

Example #4 - Input Data				
Systematic Data				
Interval Data				
Perception Thresholds				
Summary Statistics				
Year	Lower	Most Likely	Upper	Plotting Position
1820	12,526	12,527	10,000,000	0.004474

Example #4 - Input Data		
Systematic Data		
Interval Data		
Perception Thresholds		
Summary Statistics		
Start Year	End Year	Value
1820	1937	12,525

Figure 4 – Example #4 binomial censored data.

Results for this example are compared in Figure 5 below.

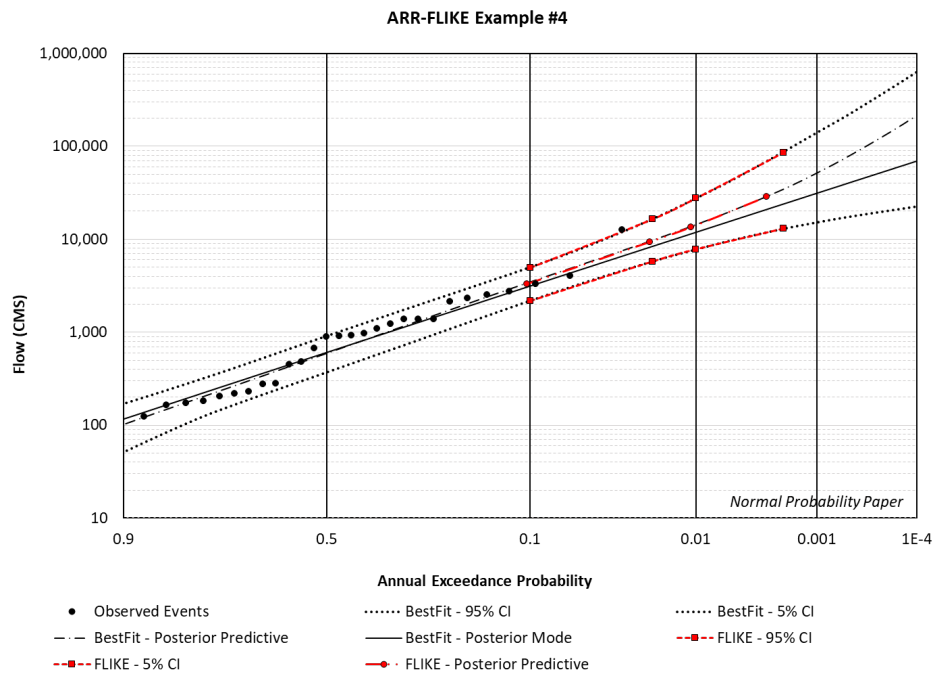


Figure 5 - Comparison Between FLIKE and RMC-BestFit (Gamma Prior for Standard Deviation) for Example #4.

Plotting Positions in RMC-BestFit

The paper suggests that RMC-BestFit does not update the graphical portrayal of results when the user changes the plotting position selection. We have been unable to replicate this issue. Users experiencing this issue, or any other issue, are encouraged to submit details to RMC.Software@usace.army.mil so that we can investigate and take appropriate action to address the issue.

RMC-BestFit supports multiple plotting position options, including Cunnane. The selected plotting position is used to portray the graphical results. The Weibull plotting position is probability unbiased making it the best choice for comparing plotting positions to a posterior predictive curve. This is why

Weibull is the default option in RMC-BestFit. The Cunnane plotting position is practically quantile unbiased for a Log-Pearson Type III distribution making it the best choice for comparing plotting positions to the posterior mode curve of a Log-Pearson Type III distribution. In practice, the choice of plotting position rarely has any impact on our understanding of the flood hazard, the associated risks, or on our investment decisions.

The preferred plotting position can be selected using the drop-down list found on the input properties window as shown in Figure 6 below. Changing this selection will also change the corresponding graphical portrayal of results.

The screenshot shows the 'Properties' window with the following fields:

- Input Data Properties:**
 - Name: Example #3 - Input Data
 - Description: Hunter River at Singleton
 - Created On: 1/10/2020 3:06:00 PM
 - Last Modified: 12/23/2023 10:30:02 AM
 - Unit Label: Peak Flow (CMS)
- Plotting Positions:**
 - Parameter: Weibull ($\alpha = 0.0$)
 - Multiple Grubbs: Blom ($\alpha = 0.375$)
 - Threshold Value: Cunnane ($\alpha = 0.40$)

The dropdown menu for the 'Parameter' field is open, showing the following options:

- Weibull ($\alpha = 0.0$)
- Median ($\alpha = 0.3175$)
- Blom ($\alpha = 0.375$)
- Cunnane ($\alpha = 0.40$)
- Gringorten ($\alpha = 0.44$)
- Hazen ($\alpha = 0.50$)

Figure 6 - Plotting Position options.

Table 2 and Figure 7 below show a comparison between the Weibull plotting positions and the Cunnane plotting positions for Example #3 in Book 3 of ARR 2019.

Table 2 - Comparison of Plotting Position results.

Flow (cms)	Plotting Position Expressed as AEP		
	FLIKE v5.306	RMC-BestFit (Cunnane)	RMC-BestFit (Weibull)
12,526	0.0192 (1 in 52.0)	0.0192 (1 in 52.0)	0.0313 (1 in 32.0)
4,052	0.0513 (1 in 19.5)	0.0513 (1 in 19.5)	0.0625 (1 in 16.0)
3,316	0.0833 (1 in 12.0)	0.0833 (1 in 12.0)	0.0938 (1 in 10.7)
2,752	0.115 (1 in 8.67)	0.115 (1 in 8.67)	0.125 (1 in 8.0)
2,536	0.147 (1 in 6.78)	0.147 (1 in 6.78)	0.156 (1 in 6.4)

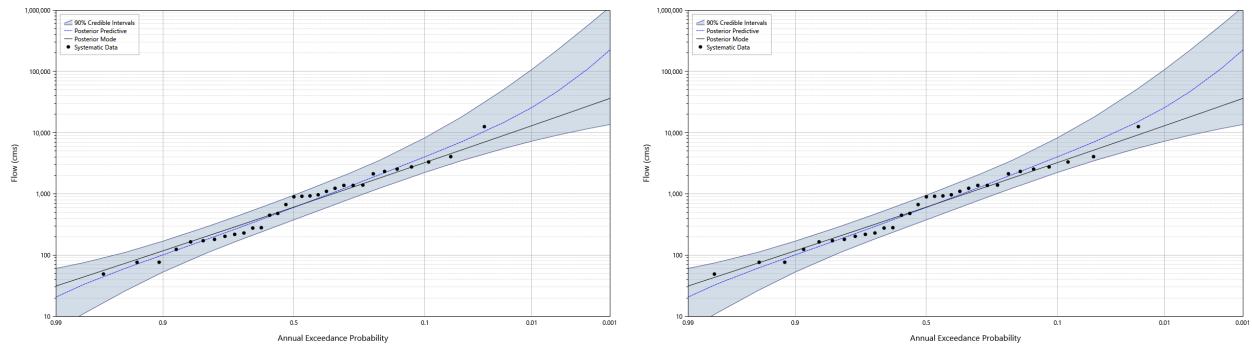


Figure 7 - Comparison between Weibull (left) and Cunnane (right) plotting positions for Example #3 in Book 3 of ARR 2019.

Credible Intervals

The paper suggests that RMC-BestFit computes credible intervals around the posterior mode curve rather than the posterior predictive curve. This is a common misconception. The credible intervals are not constructed around either curve. The credible intervals in RMC-BestFit are equivalent to the credible intervals in FLIKE. RMC-BestFit computes equal-tailed credible intervals.

In our experience, misinterpretation of the credible interval happens when we observe the posterior predictive curve approaching and eventually crossing the upper credible limit as the exceedance probability gets smaller. This behavior is correct and occurs because the distribution of exceedance probability for a given value of the random variable is symmetrical when the exceedance probability is at or near 0.5. As the exceedance probability gets smaller and smaller, the distribution of exceedance probability becomes more and more positively skewed. The positive skewness pulls the expected value toward the upper credible limit. At extremely small values of exceedance probability, the posterior predictive curve will always cross the upper credible limit eventually. Conversely, the uncertainty distribution becomes more and more negatively skewed as the value of the random variable gets smaller and smaller, pulling the expected value toward the lower credible limit. This is why the posterior predictive curve will always trend toward the upper credible limit as the exceedance probability gets smaller and smaller, and always toward the lower credible limit as the exceedance probability gets larger and larger. The inflection point will always occur at an exceedance probability at or near 0.5 depending on the distribution type.

Figure 8, Figure 9, and Figure 10 below illustrate this behavior using results from Example #3 in Book 3 of ARR 2019.

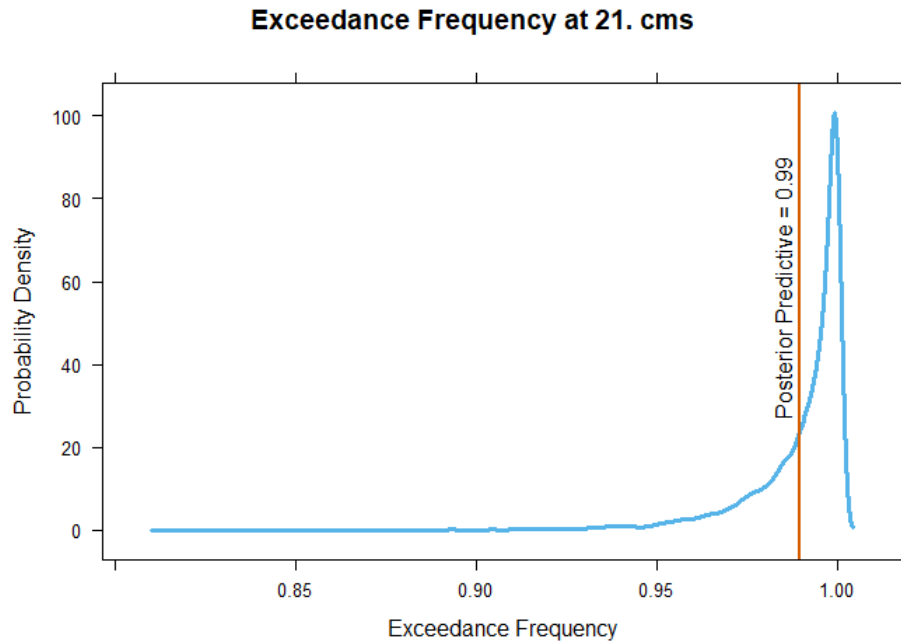


Figure 8 - An example of a negatively skewed distribution for an exceedance probability greater than 0.5 where the left tail pulls the mean of the distribution toward the lower credible limit.

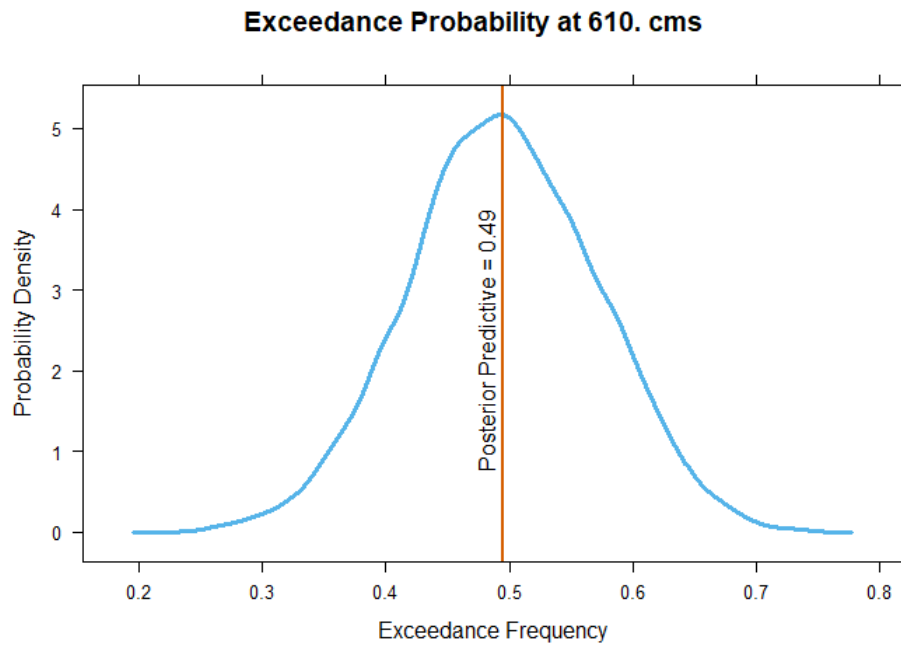


Figure 9 - An example of a symmetric distribution for an exceedance probability near 0.5 where the mean of the distribution is in the middle of the credible interval.

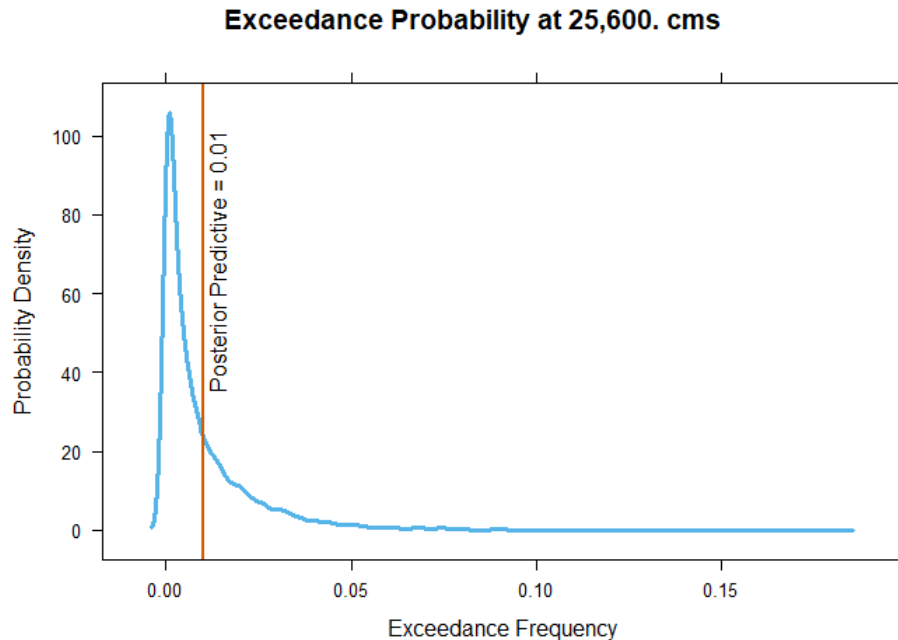


Figure 10 - An example of a positively skewed distribution for an exceedance probability less than 0.5 where the right tail pulls the mean of the distribution toward the upper credible limit.

Credible intervals are not defined or constructed around either the posterior predictive or posterior mode curves and should not be interpreted in this way. The posterior mode curve, posterior predictive curve, and credible intervals are constructed from posterior parameter sets that are sampled from the posterior parameter distributions. By default, RMC-BestFit generates 10,000 posterior parameter sets and reports results at 25 exceedance probabilities.

The general procedure for calculating the posterior mode curve is to select the posterior parameter set that has the highest (maximum) likelihood. From this parameter set, we can then compute values of the random variable (e.g., flow) at each exceedance probability using the distribution quantile function (e.g., Log-Pearson Type III). The result is equivalent to the result that we would obtain using *maximum likelihood estimation* (MLE) with uniform priors, or *maximum a posteriori* (MAP) with informative priors.

Calculation of the posterior predictive curve is a bit different because it is defined by the expected value of exceedance probability at a given value of the random variable. The general procedure for calculating the posterior predictive curve is to first calculate values of exceedance probability for each parameter set over a range of values for the random variable (e.g., flow). These values represent independent random samples from the distribution of exceedance probability at a given value of the random variable. The posterior predictive curve is defined by the expected value of this distribution. We can therefore calculate the mean exceedance probability at each value of the random variable to construct the posterior predictive curve. For convenience, we use interpolation to calculate and report the results at standard exceedance probabilities.

The general procedure for calculating the credible interval is to calculate values of the random variable for each parameter set and each exceedance probability using the quantile function. These values can be conceptualized as independent random samples from the distribution of the random variable at a

given exceedance probability. The credible interval can be obtained by calculating the 95th and 5th percentile from these values at each exceedance probability.

Respectfully submitted,

Haden Smith, P.E.

Senior Hydrologic Engineer | Risk Management Center
12596 W. Bayaud Ave. Suite 400, Lakewood, CO 80228
Office: 303-963-4575
Cell: 901-569-8480



Download RMC Software and Documentation from [GitHub](#)



Email rmc.software@usace.army.mil for software inquiries

References

- [1] L. Millard, "Probably some practical advice when using Flood Frequency Analysis utilities," in *Hydrology & Water Resources Symposium*, Sydney, Australia, 2023.
- [2] G. Kuczera, "Comprehensive at-site flood frequency analysis using Monte Carlo Bayesian inference," *Water Resources Research*, pp. 1551-1557, 1999.
- [3] C. H. Smith, "RMC-TR-2020-02 Verification of the Bayesian Estimation and Fitting Software (RMC-BestFit)," U.S. Army Corps of Engineers, Lakewood, CO, 2020.
- [4] J. Ball, M. Babister, R. Nathan, W. Weeks, E. Weinmann, M. Retallick and I. Testoni, *Australian Rainfall and Runoff: A Guide to Flood Estimation*, Commonwealth of Australia, 2019.